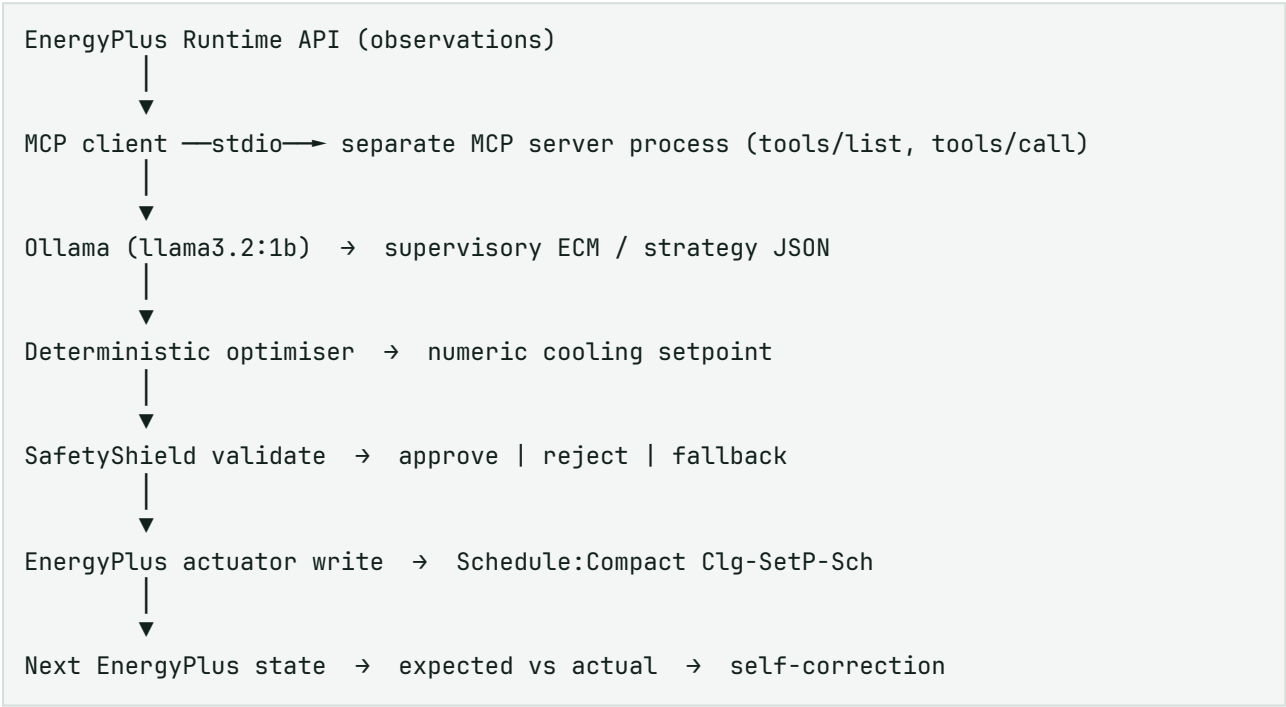


System Architecture — Eco-Loop Building Agents

TwinPilot / Eco-Loop · EnergyPlus digital building · MCP stdio · local Ollama · SafetyShield
Source of truth: docs/architecture.md

1. Closed-loop diagram



AI proposes. SafetyShield validates. EnergyPlus executes.

2. Tool-calling architecture (MCP)

Transport: stdio JSON-RPC between experiment MCP client and a *separate* MCP server OS process. Proof: mcp_pids_differ: true, traces with initialize, tools/list, tools/call.

Tool	Role
get_building_observation	Normalize EnergyPlus Runtime observation
propose_or_prepare_control_context	Structured context + prompt hint
select_energy_conservation_measure	Supervisory ECM strategy (no actuation)
validate_control_action	Advisory SafetyShield check
get_controller_constraints	Setpoint limits
record_control_decision	Audit log

get_recent_energy_history	History context
get_previous_action_outcome	Self-correction input

3. Prompt engineering

- Local Ollama llama3.2:1b with format: json
- Path C: strategy/ECM JSON only — never invents actuators
- Path B hint: setpoint JSON within 22–28°C
- Parse/HTTP failure → deterministic fallback; SafetyShield still gates writes

4. Latency management

Concern	Approach
LLM	One call per control hour; 12s timeout
Numerics	In-process deterministic optimiser
Safety	Local SafetyShield before every write
MCP	Local stdio; no cloud hop

Measured Path C obs → next hop: p50 ≈ 5.6s · p95 ≈ 6.5s (dominated by Ollama).

5. Lengthy simulation logs

- Compact KPIs in summary.json / comparison exports
- Full decisions → CSV excerpts in submission ZIP
- Protocol proof in searchable mcp-runtime-trace.jsonl
- Native EnergyPlus eplusout.* not shipped; regenerated by scripts

6. Authoritative Path A results

Metric	Baseline	Eco-Loop	Improvement
HVAC energy	13.85 kWh	13.16 kWh	4.98%
Total energy	421.51 kWh	416.00 kWh	1.31%
Peak power	19.93 kW	19.64 kW	1.47%
Comfort violations	0 h	0 h	—

7. Packages

services/{api,optimizer,simulator,agent,mcp-server} · apps/web · building-models · scripts/